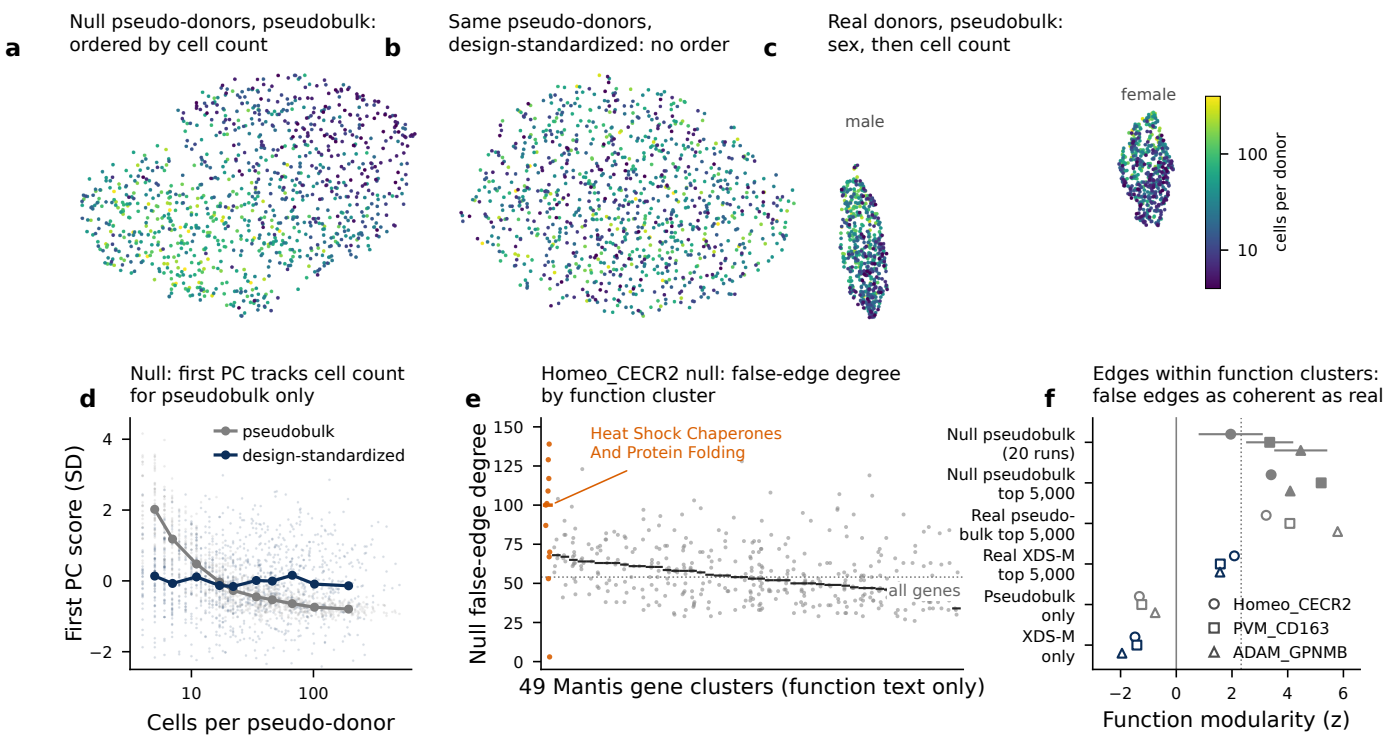

a
Null pseudo-donors, pseudobulk: ordered by cell count
b
Same pseudo-donors, design-standardized: no order
c
Real donors, pseudobulk: sex, then cell count
male
female
100
10
cells per donor
d
Null: first PC tracks cell count for pseudobulk only
pseudobulk
design-standardized
First PC score (SD)
Cells per pseudo-donor
e
Homeo_CECR2 null: false-edge degree by function cluster
Heat Shock Chaperones And Protein Folding
all genes
Null false-edge degree
49 Mantis gene clusters (function text only)
f
Edges within function clusters: false edges as coherent as real
Null pseudobulk (20 runs)
Null pseudobulk top 5,000
Real pseudo-bulk top 5,000
Real XDS-M top 5,000
Pseudobulk only
XDS-M only
Homeo_CECR2
PVM_CD163
ADAM_GPNMB
Function modularity (z)